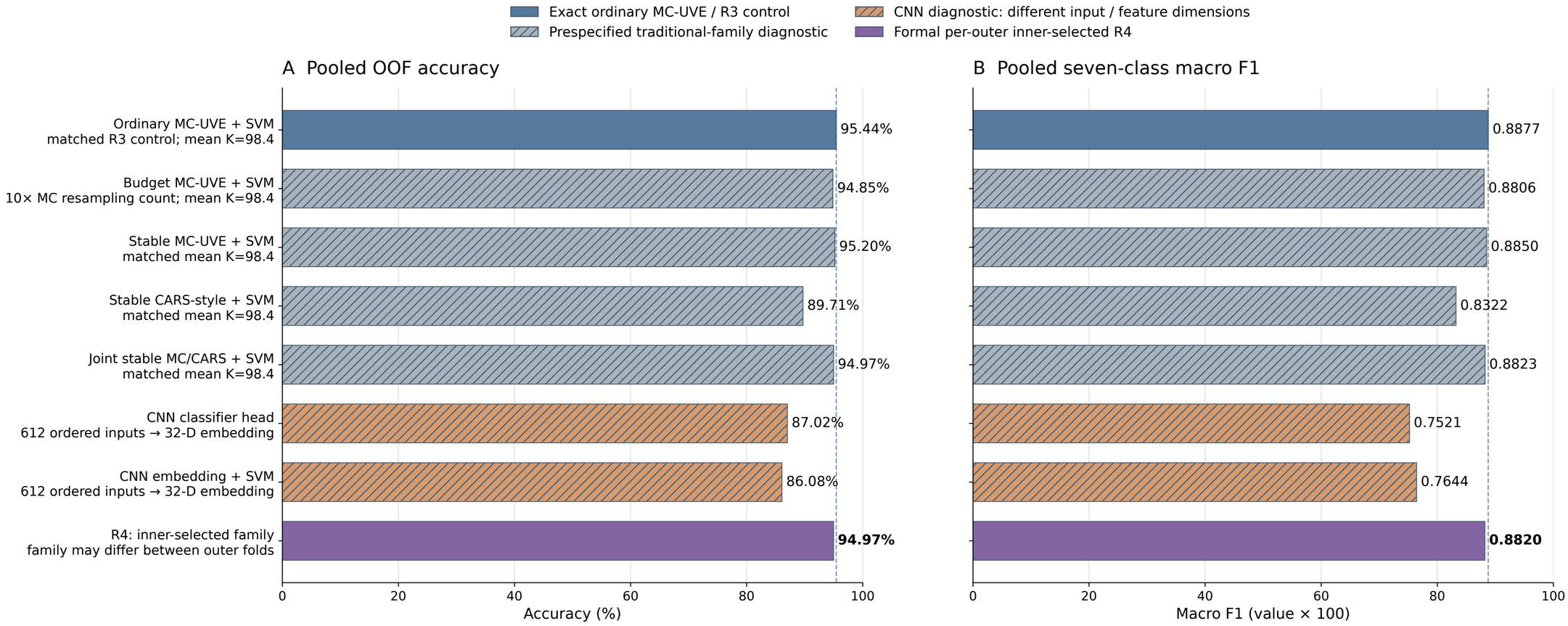
Combination-family diagnostics with matched traditional feature counts



The first five traditional selectors inherit K separately from each outer fold's R1 choice (mean K=98.4), as well as its R2 SVM and R3 preprocessing choices.

Ordinary MC-UVE is exactly the R3 control. Budget MC-UVE matches 10× resampling count, not equal time/FLOPs; stable families use source-only resamples.

CNN paths use all 612 ordered spectral inputs and a 32-dimensional embedding; they are not matched-dimension controls for the K-variable selectors.

Each diagnostic and the formal R4 pipeline predict each of the 855 specimens once out of fold. R4 family selection is inside each outer training set.

SVM heads use standard SVC vote; CNN head uses argmax. Same 5 outer / 2 inner folds, pooled Macro F1. Historical single-batch data; no independent batches or CIs.

Diagnostics do not replace the inner-selected family after outer scoring. Do not rank these results against prior 171-holdout scores; both axes start at zero.